UF UNIVERSITY of FLORIDA
Herbert Wertheim College of Engineering

CIS4XXX

Exercise: COVID Data Visualizer

Overview

This project will provide students practice working with **Django** framework, data visualization in Python by building models and data queries to visualize large public datasets on COVID-19 using different chart types.

Requirements

Each student will develop a data visualization application using Django web framework. Students will write two files to submit: a URL file as the entry point (`urls.py`), a file containing the view Python function (`views.py`) that takes a `Web request` and returns a `Web response`. This will require creating a `data model` that can be accessed in the `views.py` to retrieve the data for creating visualizations. The Python module `views.py` provides a mapping between the URL path to Python functions for different visualizations in `views.py`. When required, create a form class to take user input in file `forms.py`.

Driver Application

The `urls.py` files should include five different URLs corresponding to the questions listed below:

1) Data access to `.csv` file provided with the assignment on covid-19 data
2) Print out a Welcome message to the console.

Views Module

The following components of `views.py` correspond to different questions for which students should create visualizations using `Chart.js` from the templates provided in the **templates** folder. The visualizations should be mapped to the given URL path and the **names** of the arrays returned in the render function must match exactly those given.

Print out the `resulting data` of each function using the `print_results()` method.

Required Methods

`us_cases()`

This method generates a `double line graph` visualization that plots the total and new number of COVID cases in the United States where data is available (no null values) over time. This visualization should be mapped to the path **`"/us_cases"`**.

Return:

`'labels'`: the dates for the `data`

`Data`

`'total_cases'`: total number of cases for those dates

`'new_cases'`: new number of cases for those dates

avg_cases_and_deaths ()

This method generates a stacked bar graph data visualization to show: For which month did each country had the highest on average daily new cases along with how many total deaths happened in this month. Plot top 5 countries with the most new cases. This will be mapped to the path **"/avg_cases_and_deaths"**.

Return:

'labels': country names

Data

'new_cases': number of new cases that month (in ten thousands for scaling)

'new_deaths': number of new tests that month.

top5_radar()

This method generates a radar chart graph visualization to compare data of the top five countries with the most total cases. This will be mapped to the path **"/top5_radar"** .

Return:

'labels': country names

Data

'cases': total cases for each country (in ten thousands)

'deaths': total deaths for each country (in thousands)

'cases_per_million': cases per million * 100 for each country (for scaling)

'deaths_per_million': deaths per million for each country

max_diff()

This method generates a bar graph visualization to visualize difference in the total number of new cases in the top 5 countries with most total cases before (including May) and after May (starting June). This will be mapped to the path **"/max_diff"**.

Return:

'labels': country names

Data

'differences': the difference between total cases before and after May (absolute values)

positive_rates()

This method will ask users to input a month in numeric format (ex. March = 3). Then, take the top 5 countries with the highest total cases and find the average positive rate in those countries for the specified month. Plot these rates in a pie chart. This will be mapped to the path **"/positive_rates"**.

You will need to create **forms.py** and a textfield to get this input

Failure messages for input:

Non-numeric values

Ex. "A" -> Print: "Error: Invalid Input"

Months greater than 12 (December)

Ex. "13" -> Print: "Error: Out of Range"

Months less than 1 (January)

Ex. "0" -> Print: "Error: Out of Range"

Return:

'labels': country names

Data

'positive_rates': average positive rate in that month for each country (in percents ie. * 100)

`print_results()`
This method accepts an `array` of the resulting dataset from one of the visualizations and logs the `data` to the console.

Submissions

Please submit only and exactly these files:

Files:	views.py, urls.py, forms.py
Method:	Submit on Canvas